

SC-SCOPE endpoint: per-transfer quartic-difference and the joint inequality (honest negative)

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

Operator directive: drive the SC-SCOPE critical path to closure. With M-ENDPOINT resolved (sunset axis positive) the two remaining inputs are GHAT4-PERTRANSFER (the per-transfer quartic-difference form factor) and the joint second+third-order endpoint inequality. This note evaluates both and reports the HONEST verdict: the endpoint joint does NOT close under current bounds. Scope: the $I = 2 \times 10^{-3}$ endpoint at the production anchor; estimate-grade absolute normalisation, rigorous convention-free shape factors.

2. GHAT4-PERTRANSFER: the per-transfer quartic-difference form factor

The quartic-difference channel inflates the certified second-order budget by $R(t) = 12(5v/2)^2 \lambda'^{-2} \widehat{G^4}(t) 4(1 - a_0)/J(t)$, with $\widehat{G^4} = G * G * G * G = J * J$. The sup-kernel estimate paired the Young ceiling $\widehat{G^4} \leq J(0)M^2$ with the floor $J(t) \geq J(2q_0)$ SIMULTANEOUSLY, giving $R_{\text{sup}} \approx 1.59$ ($\Rightarrow$ inflation $\times 2.6$, endpoint $\times 1.0$ marginal). That pairing is incompatible. The convention-free shape factor

$$\Phi(t) = \frac{\widehat{G^4}(t)}{\widehat{G^4}(0)} \cdot \frac{J(0)}{J(t)}$$

(all measure prefactors and the J -normalisation factor-2 cancel) was evaluated on the realized chords $t \in [t_{\min}, 2q_0]$ with $\widehat{G^4}(t) = (J * J)(t)$ computed by a converged convolution quadrature (grid-stable to 0.00%). The maximum is at the antipodal transfer $t = 2q_0$:

$$\max_t \Phi(t) = 1.77 \ll \Phi_{\text{sup}} = J(0)/J(2q_0) = 2.77, \quad \text{reduction} = \frac{\max_t \Phi}{\Phi_{\text{sup}}} = 0.64.$$

The incompatible pairing is confirmed (the realized ratio is well below the sup), but the reduction is only 0.64: $\widehat{G^4}$ is broad, falling only to $0.64 \widehat{G^4}(0)$ at $t = 2q_0$. Hence the firmed per-transfer ratio is

$$R_{\text{max}} = R_{\text{sup}} \cdot 0.64 \approx 1.02,$$

which is ≥ 1 – precisely the condition the quartic-difference note pre-registered as triggering the joint endpoint re-derivation. Taken ALONE the quartic-difference endpoint is $\times 2.6/(1 + R_{\text{max}}) = \times 1.29 > 1$; R_{sup} is the cited convention-bearing estimate (not recomputed here).

3. The joint endpoint inequality (the honest negative)

Each third-order channel clears the endpoint margin INDIVIDUALLY – second-order $\times 2.60$, sunset $\times 1.13$ (dressed, M-ENDPOINT), quartic-difference $\times 1.29$ (per-transfer), tadpole 0 (U6). But all costs draw on the SAME layer margin. With $C_2 = \text{MARGIN}/\rho$, composed = $\text{MARGIN}(1 - 1/\rho)$, $C_{\text{sunset}} = \text{composed}/1.13$, $C_{\text{quartic}} = R_{\text{max}}C_2$, $C_{\text{tadpole}} = 0$ ($\rho = 2.6$, $\text{MARGIN} = 0.00432$):

$$\frac{\text{MARGIN}}{C_2 + C_{\text{sunset}} + C_{\text{quartic}}} = \times 0.757 < 1.$$

The summed second+third-order cost exceeds the layer margin by $\times 1.32$. The SC-SCOPE all-orders endpoint therefore does NOT close under the current (individually positive) bounds. This is an honest negative, registered in negative-results.

4. What closure would require

The individual-positivity / joint-failure gap ($\times 1.32$) is the signature of a bound that is still loose at the JOINT level: the sunset bound is largest at SMALL transfers (it rides $J(0)M$), the quartic-difference is largest at LARGE transfers (Φ peaks at $t = 2q_0$), and the second-order is spread across the chord set. A JOINT incompatible-pairing argument – bounding $\max_t [c_{\text{sunset}}(t) + c_{\text{quartic}}(t)]$ rather than $\max_t c_{\text{sunset}} + \max_t c_{\text{quartic}}$ – would recover up to the $\times 1.32$ deficit, exactly as the single-channel per-transfer argument recovered the sup-grade crudeness one level down. Sharper per-transfer sunset and quartic bounds (both currently inherit one sup-grade factor) are the alternative. Either is a genuine new derivation, registered as the next SC-SCOPE step; neither is performed here.

5. Impact: SC-SCOPE OPEN, selection unaffected

SC-SCOPE stays OPEN at the endpoint. This does NOT weaken B1's T6 selection, which holds at matched second-cumulant order BY THE SC-SCOPE HYPOTHESIS: SC-SCOPE is a named hypothesis of B1 T6, not a proved input, so its remaining-open status is already priced into the tier. What is now precisely characterised is the all-orders ENDPOINT obstruction: a joint $\times 1.32$ deficit with an explicit path to close it. M-ENDPOINT (resolved) and the GHAT4 per-transfer reduction (0.64) are durable advances toward that joint argument.

6. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3, 6 code-discipline)

- (α) “*You are over-pessimistic: the channels never all peak at one transfer, so the additive joint is too crude.*” CORRECT, and stated in §4: the additive joint is an UPPER bound on the cost; the honest negative is that the CURRENT bounds do not close, with the joint incompatible-pairing argument named as the route. The note does not claim the endpoint FAILS physically, only that it is not yet PROVED.
- (β) “ *$R_{\text{max}} \approx 1.02$ leans on the cited $R_{\text{sup}} = 1.59$, an unverified estimate.*” VALID: the convention-free shape reduction 0.64 is rigorous; the absolute R_{max} inherits R_{sup} . If R_{sup} is loose the quartic-difference improves, but the sunset contribution alone ($C_{\text{sunset}} = \text{composed}/1.13$) plus

C_2 already gives $\text{MARGIN}/(C_2 + C_{\text{sunset}}) = \times 1.06$ – so the endpoint is tight even before the quartic-difference. The negative is robust to R_{sup} .

(γ) “Does this threaten B1 T6?” DISMISSED: SC-SCOPE is a NAMED hypothesis of B1 T6; an open hypothesis is the tier’s explicit condition, not a refutation.

Quantitative sanity check (mandatory): magnitude – the convolution $\widehat{G^4}(0)$ is grid-stable to 0.00%; ratio – $J(0)/J(2q_0) = 2.77$ reproduces the note’s 2.79; limit – as $t \rightarrow 0$, $\Phi \rightarrow 1$; sign-direction – all third-order channels are competitor-helping (cost, not credit), so they add against the margin, the conservative direction; consistency – each channel > 1 individually yet joint < 1 is the expected signature of additive bounds on a shared budget.

7. Result footer

Result ID:	B5-BEYOND-LAYER-BOUND / SC-SCOPE endpoint joint (U15/U16)
Precise statement:	GHAT4-PERTRANSFER: per-transfer $\text{Ghat4}(t)=(J*J)(t)$ on the realized chords; convention-free shape reduction $\max \Phi / \Phi_{\text{sup}} = 0.64$ (Ghat4 broad), $R_{\text{max}} \sim 1.02$. JOINT endpoint: $\text{MARGIN} / (C_2 + C_{\text{sunset}} + C_{\text{quartic}}) = x0.757 < 1$ -- the individually-positive 2nd/sunset/quartic channels jointly over-consume by $\times 1.32$. SC-SCOPE endpoint does NOT close under current bounds.
Scope:	$I=2e-3$ endpoint, production anchor; shape factors rigorous, absolute R_{sup} cited (estimate).
Dependencies:	sectorb_common (J_of_t, margin_of); scscope-mendpoint-evaluation v1.1 (sunset $\times 1.13$); quartic-difference-channel v1.0 ($R_{\text{sup}}=1.59$ cited).
Evidence grade:	EXECUTED (scscope_ghat4_pertransfer.py 7/7; scscope_joint_endpoint.py 5/5); honest negative.
Reproduction command:	python codes/vacuum/scscope_ghat4_pertransfer.py && python codes/vacuum/scscope_joint_endpoint.py
Expected output:	reduction 0.64, $R_{\text{max}} 1.02$; joint endpoint $x0.757 < 1$.
Falsification gate:	A joint incompatible-pairing bound, or sharper per-transfer sunset/quartic bounds, giving $\text{MARGIN}/\text{sum} > 1$ (would CLOSE the endpoint and is the registered next step).
Tier before / after:	No tier/gate flip. GHAT4-PERTRANSFER: per-transfer evaluated (shape reduction 0.64), endpoint NOT resolved alone. SC-SCOPE: OPEN (endpoint joint $x0.757$). B1 T6 unchanged (SC-SCOPE is a named hypothesis).
No-overclaim statement:	Honest negative: SC-SCOPE does NOT close. Each channel is individually positive; the joint is not. The selection is unaffected (second-cumulant by hypothesis). Absolute R_{sup} is a cited estimate; the shape reduction and the joint arithmetic are exact given inputs.
Next required action:	Joint incompatible-pairing argument (bound $\max_t [c_{\text{sunset}}(t) + c_{\text{quartic}}(t)]$) OR sharper per-transfer sunset/quartic bounds; then re-test the joint endpoint.